

DOES MACHINE LEARNING BEAT THE CREDIT SCORECARD? DISCRIMINATION, CALIBRATION AND PROFIT IN PROBABILITY-OF-DEFAULT MODELLING

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ABSTRACT

Banks deploy logistic-regression scorecards for probability-of-default (PD) modelling, while the machine-learning literature reports that gradient-boosted trees dominate them. That literature scores models almost exclusively by discrimination (AUC), whereas a lender also needs calibrated probabilities and profitable accept/reject decisions. This project tests whether the ML advantage survives all three criteria. An industry-style weight-of-evidence scorecard and three challengers (XGBoost, LightGBM, a multilayer perceptron) were built on 621,022 Lending Club loans under a strict out-of-time protocol using origination-time features only. XGBoost beats the scorecard by 0.85 AUC points (DeLong $p < 10^{-54}$), keeps the edge after identical recalibration, and earns 2.1% more profit at its optimal cutoff; the MLP is significantly *worse* than the scorecard, and neither tree ensemble matches the scorecard’s calibration. An ablation withholding the platform’s own risk pricing shows the modest margin to be a property of the information set rather than of the models: the scorecard then degrades almost twice as fast as the trees and XGBoost’s lead nearly triples, to 2.39 points. Boosted trees earn their complexity, but the forty-year-old scorecard covers most of the distance.

Index Terms: Credit scoring, probability of default, calibration, XGBoost, weight of evidence, profit-based evaluation

1. INTRODUCTION

When a bank grants a loan, the decision rests on one model output: the applicant’s probability of default (PD). PD models set rates and limits, drive the accept/reject decision, and under the Basel accords and IFRS 9 determine regulatory capital and loan-loss provisions [1]. Because capital scales with estimated PDs, these are among the highest-stakes statistical models in finance, and regulators require them to be accurate, stable and explainable. The industry standard is old technology: a logistic regression on weight-of-evidence (WoE) transformed features, packaged as an additive point scorecard [3, 2].

A large benchmarking literature reports that gradient-boosted trees outperform logistic regression on credit data [4, 5], but states that evidence almost entirely as discrimination, the ability to rank bad borrowers above good ones, summarized by AUC. Discrimination is not what a lender consumes. Provisioning needs the PD itself to be right on average (calibration), and lending needs the model to make money at an operating cutoff (profit). A model can gain AUC while producing badly miscalibrated probabilities.

This project puts the question to a controlled test. One pipeline builds a scorecard and three challengers on identical

origination-time information, evaluates them out of time, applies identical recalibration to every model, and converts predictions into realized profit, so any claimed advantage must survive calibration and decision analysis rather than an AUC comparison alone. Section 2 states the problem, Section 3 the method, Section 4 the experiments and results, Section 5 discusses, and Section 6 concludes.

2. PROBLEM STATEMENT AND OBJECTIVES

Let $(\mathbf{x}_i, y_i)$ describe loan i , where $\mathbf{x}_i$ holds only information available on the issue date and $y_i \in \{0, 1\}$ marks eventual charge-off. The task is to produce $\hat{p}_i = \Pr(y_i = 1 | \mathbf{x}_i)$ such that risky applicants rank above safe ones, $\hat{p}_i$ is numerically trustworthy, and an accept/reject rule built on it is profitable.

Two features of the problem shape the design. First, *leakage is the dominant failure mode*: a raw servicing extract contains dozens of post-origination fields that trivially predict the outcome, so the feature set must be an explicit origination-time whitelist. Second, *deployment is out of time*: a model trained on past cohorts scores future applicants under a shifted economy, so a random split flatters every model and splits must follow the calendar.

The objectives were to implement an honest industry-standard WoE scorecard; to implement three ML challengers on identical information; to evaluate discrimination, calibration before and after identical recalibration, and realized profit, each with significance testing; to quantify how much of the outcome depends on the platform’s own risk pricing; and to verify robustness on an independent dataset with a reproducible codebase. The scope is deliberately narrow: one consumer-credit market, 36-month loans only so every outcome is observed, and a fixed loss-given-default assumption.

3. METHODOLOGY

Workflow. The raw export is reduced to the origination-time whitelist; loans are split by issue date into train, validation and test vintages; the scorecard and challengers are fit on train with validation used for early stopping; every model’s raw PDs are recalibrated on the validation vintage; raw and recalibrated PDs are scored on the test vintage with DeLong tests and bootstrap intervals; and each model’s PDs drive an accept/reject sweep scored on the test loans’ actual cash flows. ML models are retrained across five seeds.

Scorecard. Numeric features are quantile-binned on training data (missing values form their own bin) and categorical levels are pooled when rare. Bin j receives the

weight of evidence

$$\text{WoE}_j = \ln\left(\frac{g_j/g}{b_j/b}\right), \quad (1)$$

with g_j, b_j the goods and bads in the bin and g, b their totals, counts smoothed by 0.5. Features are screened by information value, $\text{IV} = \sum_j (g_j/g - b_j/b) \text{WoE}_j$, keeping $\text{IV} \geq 0.02$ [3]. A logistic regression on the WoE features yields the PD, also expressed as additive points scaled so that 600 points correspond to 19:1 good:bad odds and 20 points double the odds. Every input enters through a monotone-ish transform and the model is a readable sum of points.

Challengers. XGBoost [6] and LightGBM [7] are boosted tree ensembles (2,000 trees maximum, learning rate 0.05, depth 5 and 31 leaves respectively, row and column subsampling 0.8, early-stopped on validation AUC); the MLP is a two-layer network (64 and 32 units, ReLU, Adam, early stopping) on standardized inputs. All three consume one design matrix, median-imputed numerics plus one-hot categoricals fit on train only, so differences come from the learner rather than the preprocessing.

Calibration. A PD model is calibrated if, among loans assigned $\hat{p} \approx q$, a fraction q defaults. Platt scaling [8] refits a one-parameter logistic on the raw model logit and isotonic regression [9] fits a monotone step function; both are fit on the validation vintage and applied to test scores, mimicking a bank recalibrating on its most recent observed cohort. Calibration is measured by the Brier score and by the expected calibration error, the mean absolute gap between predicted PD and observed default rate over ten equal-frequency bins [10].

Profit and significance. The lender accepts every applicant with $\hat{p}_i \leq c$ and c is swept over $(0, 1)$. A repaid 36-month loan earns $36 \cdot \text{installment}_i - \text{principal}_i$ (undiscounted) and a charged-off loan loses $\text{LGD} \times \text{principal}_i$ with $\text{LGD} = 0.65$; profit is reported per 1,000 applicants. AUC differences on the shared test set are tested by DeLong’s method for correlated ROC curves [11, 12], with percentile bootstrap intervals over 1,000 resamples.

4. EXPERIMENTS AND RESULTS

4.1. Data

Lending Club accepted personal loans, public export 2007–2018Q4 [14]: 2,260,701 loans and 151 columns. Restricting to 36-month loans issued 2007–2015 guarantees every loan matured before the data ends, so no outcome is right-censored; keeping only resolved outcomes leaves **621,022 loans** issued June 2007 to December 2015, of which **13.95%** were charged off.

These are small unsecured consumer loans. Amounts run from \$500 to the platform ceiling of \$35,000, median \$10,000 and mean \$12,481, with a median monthly installment of \$342. Interest rates span 5.3% to 29.0%, mean near 12%. Borrowers report a median income of \$60,000, a median debt-to-income ratio of 17% and a median 15 years of credit history. Purposes are dominated by refinancing (56.6% debt consolidation, 24.2% credit card); housing splits 46.4% mortgage, 43.4% renting, 10.2% ownership. Grades concentrate in the middle of the platform’s scale (A

Table 1. Test vintage (283,026 loans issued 2015). AUC and KS on raw scores, Brier and ECE after isotonic recalibration, with raw ECE for comparison; ML entries are means over five seeds (AUC std ≤ 0.006). ΔAUC is against the scorecard, with the DeLong p -value.

Model	AUC	KS	Brier (iso)	ECE raw / iso	ΔAUC (DeLong)
Scorecard	0.6767	0.2570	0.1209	.025/.0175	—
XGBoost	0.6850	0.2701	0.1203	.034/.0214	+ .0085***
LightGBM	0.6843	0.2693	0.1204	.034/.0218	+ .0072***
MLP	0.6716	0.2542	0.1212	.028/ 0.0172	− .0017*

*** $p < 10^{-48}$ * $p = 0.024$

23.0%, B 34.1%, C 26.2%, D 12.3%, E–G 4.3%). Lending Club’s own acceptance policy is visible in the scores: the minimum FICO is 612 and the median only 692, a narrow band that limits what any model can extract from that field.

Features and leakage. The whitelist retains 23 origination-time fields: loan terms (`loan_amnt`, `term`, `int_rate`, `installment`, `grade`, `sub_grade`); applicant profile (`emp_length`, `annual_inc`, `home_ownership`, `dti`, `purpose`, `verification_status`, `application_type`); and the bureau snapshot (`fico_range_low/high`, `earliest_cr_line`, `delinq_2yrs`, `inq_last_6mths`, `open_acc`, `pub_rec`, `revol_bal`, `revol_util`, `total_acc`, `mort_acc`, `pub_rec_bankruptcies`). Two engineered fields, the FICO band midpoint and credit-history length, bring the modelling set to 17 numeric and 6 categorical features, of which the IV screen retains 11.

The 128 discarded columns are dominated by post-origination fields that exist only after money has moved: `total_pymnt`, `total_rec_prncp`, `recoveries`, `collection_recovery_fee`, `last_pymnt_amnt` and `last_fico_range_high` are representative. The criterion is temporal, not statistical, and the danger is not subtle: `total_rec_prncp` equals the loan amount for every repaid loan and therefore encodes the target outright. The remainder are identifiers (`id`, `member_id`, `url`), free text (`emp_title`, `desc`) and fields that are almost entirely empty.

Split and environment. Train: 175,426 loans issued 2007–2013 (12.6% default). Validation: 162,570 issued 2014 (13.7%). Test: 283,026 issued 2015 (14.9%). The rising default rate is itself part of the experiment, being exactly the drift a deployed model faces. Robustness uses the UCI Taiwan credit-card set [13], 30,000 clients with a stratified 60/20/20 split. Windows 11, CPU only, Python 3.13; all seeds fixed; the full study runs in about 18 minutes.

4.2. Results

Table 1 reports the test vintage. XGBoost leads the scorecard by 0.85 AUC points and LightGBM by 0.72, both overwhelmingly significant because 283,026 test loans give enormous power; the MLP is significantly *worse*. Isotonic and Platt recalibration performed alike, so only isotonic is shown.

Calibration tells a different story. Figure 1 shows every raw curve lying above the diagonal: the 2015 vintage defaulted more than the training years, so all models under-

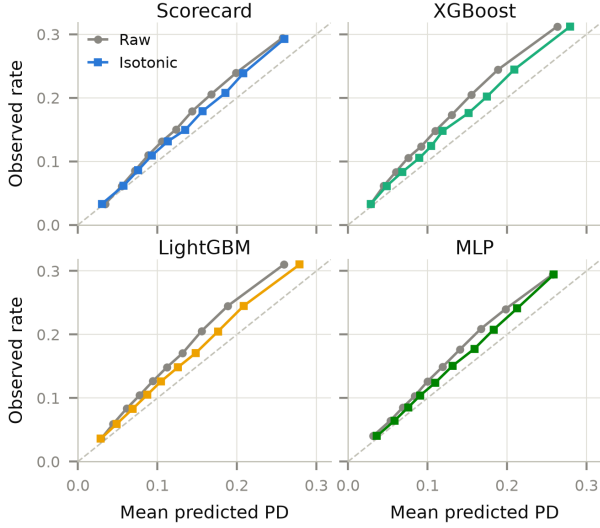


Fig. 1. Reliability on the test vintage, ten equal-frequency bins, raw (grey) versus isotonic (colour). Every raw curve sits above the diagonal because the test vintage defaulted more than the training years.

Table 2. Best operating points on the profit curves, isotonic-calibrated PDs. Profit in USD per 1,000 applicants.

Model	Cutoff	Accept	Profit
Accept all	1.00	100.0%	705,171
Scorecard	0.24	93.3%	713,772
MLP	0.25	94.0%	714,401
LightGBM	0.25	93.7%	725,927
XGBoost	0.23	90.7%	728,517

predict PD. Raw tree outputs are the worst offenders (ECE 0.034 against the scorecard’s 0.025), as expected for margin-maximizing ensembles [10]. Isotonic recalibration removes most of the bias but cannot anticipate drift not yet visible in 2014, and neither tree ensemble matches the scorecard afterwards (0.0214 and 0.0218 against 0.0175); the trees keep the better Brier score because Brier also rewards ranking.

Profit concentrates the AUC gap at the cutoff (Fig. 2, Table 2). Accepting everyone already earns \$705 per applicant because interest income dominates this portfolio, so a PD model adds value only by pruning the riskiest tail. The scorecard’s best policy rejects 6.7% of applicants for a 1.2% gain; XGBoost rejects 9.3% for 3.3%, worth about \$14,700 per 1,000 applicants over the scorecard, or 2.1%. On the Taiwan data the ranking reproduces: XGBoost leads the scorecard by 1.50 AUC points ($p = 2.8 \times 10^{-6}$) and the MLP again falls below it.

4.3. Ablation: Withholding the Platform’s Pricing

grade, sub_grade and int_rate supply 35.8% of XGBoost’s total gain and 42.5% of LightGBM’s, and are the scorecard’s three highest information values. They are not three measurements but one, namely Lending Club’s own risk assessment: within a sub-grade the rate varies with a standard deviation of 0.33 percentage points against an overall spread of 3.69. They are legitimate inputs, published before a loan is funded and genuinely observed by an investor,

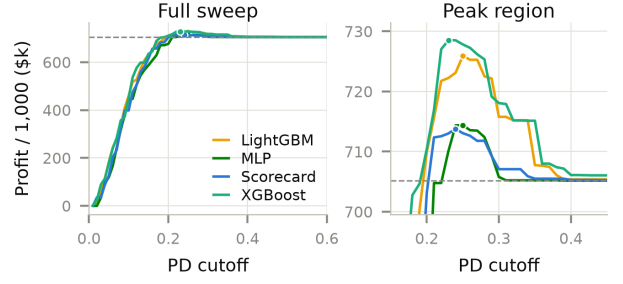


Fig. 2. Realized profit per 1,000 applicants against PD cutoff, with best operating points marked and the accept-everyone benchmark dashed. Right: the peak region, where the boosted trees separate.

Table 3. Ablation. Test-vintage AUC, mean over five seeds, when the platform’s pricing is withheld.

Model	Baseline	Var. A	Var. B	ΔB
XGBoost	0.6850	0.6678	0.6637	−0.0214
LightGBM	0.6843	0.6664	0.6622	−0.0220
MLP	0.6716	0.6488	0.6428	−0.0289
Scorecard	0.6767	0.6397	0.6397	−0.0371

XGBoost’s lead over the scorecard

AUC points	+0.83	+2.80	+2.39
Profit/1k	\$14,744	\$17,068	\$15,404

so they are not leakage. They do narrow the question, from whether ML assesses credit risk better than a scorecard to whether it re-ranks a population the platform has already ranked. The protocol was therefore rerun without them, in two variants:

- **Variant A** drops grade, sub_grade, int_rate, leaving 20 features.
- **Variant B** additionally drops installment, leaving 19.

Variant B exists because the instalment is a deterministic function of amount, term and rate; with the term fixed at 36 months the instalment per dollar borrowed correlates with the rate at $\rho = 0.998$, so under Variant A pricing remains available through that proxy. Variant B is the honest ablation and Variant A measures how far each model exploits the proxy.

Table 3 gives the outcome. **The pricing features were suppressing the ML advantage, not creating it.** Every model degrades, but unequally: the scorecard loses 3.71 AUC points against the trees’ 2.15, so XGBoost’s lead grows from +0.83 to +2.39 points under Variant B, with DeLong p underflowing to zero and a bootstrap interval of $[+0.0222, +0.0255]$; its profit advantage widens to \$15,404 per 1,000. The modest margin in the main result is thus a property of the enriched information set: once models must build a risk ordering from bureau data alone, the interaction structure that trees capture and an additive scorecard cannot become materially more valuable.

The trees used the instalment proxy and the scorecard did not. The scorecard scores 0.6397 under both variants, identical to four decimals, because installment has an information value of 0.0023 and had already been eliminated

by the IV screen; both tree ensembles lose a further 0.4 points when it is removed. The univariate screen that looks like the scorecard’s most old-fashioned step is what protected it from a proxy the flexible models consumed without complaint. The MLP remains unreliable rather than merely weaker: its seed standard deviation of 0.0056 is an order of magnitude larger than the trees’, large enough that under Variant A its five-seed mean and its reference seed disagree about whether it beats the scorecard.

5. DISCUSSION

The advantage is real but belongs to the trees. The DeLong p -values are astronomically small because of test-set size, yet the economic gap of 0.85 AUC points is an order of magnitude smaller than the 8 to 15 points sometimes quoted from random-split studies, and the MLP loses outright, echoing [5] on deep learning for tabular credit data. Two mechanisms compress it: the information set already embeds Lending Club’s underwriting, which Section 4.3 quantifies; and the out-of-time protocol denies flexible models the flattery of a random split in which train and test share one macroeconomic regime.

Calibration is a separate axis, and the scorecard holds it. Raw tree outputs are visibly miscalibrated while the logistic scorecard is a probability model by construction, and identical recalibration narrows but does not erase the difference. The reliability diagrams expose the deeper issue: all raw curves sit above the diagonal because the test vintage defaulted more than the training years, and a calibrator fitted on 2014 corrects only the drift already visible in 2014. No within-model fix can anticipate a level shift that has not happened yet, which is why regulators require ongoing monitoring rather than one-time calibration.

Profit concentrates the gap at the cutoff. Accepting everyone is already profitable, so the model adds value only by pruning the riskiest tail; switching from scorecard to XGBoost is worth 2.1%, an edge whose value scales linearly with volume, which is why large lenders adopt gradient boosting and small ones often rationally do not. The scorecard’s WoE pipeline cost discrimination but bought near-calibration and full point-by-point readability, an attractive bargain under model-governance constraints.

Limitations. The data contain only *accepted* loans, so every model inherits Lending Club’s selection; reject inference remains an open problem [2]. The profit model is undiscounted with a fixed LGD and ignores partial recoveries and prepayment. One market and one test vintage: 2015 was benign, and a recession vintage could reorder the calibration findings. The scorecard used equal-frequency bins without monotonicity constraints, so a production build with expert binning would likely close part of the remaining gap.

6. CONCLUSION AND FUTURE WORK

Under a leak-proof out-of-time protocol on 621,022 loans, gradient-boosted trees beat the industry-standard scorecard on discrimination, on Brier score after identical recalibration, and on realized profit, and the ranking reproduced on an independent dataset. The answer is a qualified *yes*: statistically unambiguous, economically modest at 2.1% profit, delivered only by the boosted trees, and achieved without

matching the scorecard’s calibration. The margin is also conditional on the information set, nearly tripling once the platform’s own pricing is withheld, so the headline figure understates what ML offers a lender underwriting from bureau data directly.

The main learning outcomes were methodological. Leakage control was the highest-leverage step in the project, and any mistake there would have produced impressive, meaningless results. Evaluation criteria are not interchangeable: the ranking changed with the question asked, so “which model is best” has no answer until the deployment question is fixed. And vintage drift is not a nuisance but the phenomenon itself. Future work: reject inference; survival models for the timing of default; discounted cash flows with empirical recoveries; monotonicity-constrained boosting as a regulator-friendly middle ground; and stress evaluation on a recession vintage.

7. ARTIFACTS

Code, results and both reports: <https://github.com/iitgF/T8>, containing the modular `creditrisk` package and the run script reproducing every number above. LaTeX source: <https://www.overleaf.com/project/6a5345413275caec079cf796>. YouTube: <https://youtu.be/RNXtupZw2UE>

Attribution note: AI-assisted tooling (Claude Code) was used for code refactoring, optimization and drafting support. All modelling decisions, verification of results and the final text are the author’s responsibility, and all reported numbers were reproduced end-to-end from the committed code.

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